CERTIFICATE

*This is to certify that Mr./Ms.*____

of ______________________________ *Class, Roll No.* ____________________

Exam No. _____________________ *has satisfactorily completed his / her term*

work in ____ *for*

the term ending _________________ *in 20*___ */ 20*___.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHANGA – 388 421

Date:

Sign of the Faculty *Head of the Department*

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty of Technology and Engineering
Manubhai Shivabhai Patel Department of Civil Engineering

CL433 ENVIRONMENTAL POLLUTION AND CONTROL

INDEX

Date:

1.

INTRODUCTION TO SAMPLING AND STANDARDS FOR WATER AND WASTEWATER ANALYSIS

Sampling:

Introduction

A sample is a unit portion of water or wastewater drawn by scientific method for getting information regarding characteristics of water and wastewater.

Sampling is used every day at water and wastewater treatment plants to determine the characteristics of the water. Sampling may be used to test the finished water for regulatory purposes - to ensure that the treatment plant is treating the water in compliance with regulations.

The sample must be collected in such a manner that nothing is added or lost in the portion taken, and so that no change occurs in the sample's characteristics between the time the sample is collected and the time the laboratory test is performed. Since the value of any laboratory result depends on the quality of the sample taken, bad sampling can lead to laboratory results which are misleading and worse than no results at all.

Sampling Guidelines

Safety

Sampling safety is similar to general laboratory safety. When dealing with wastewater of any sort, the operator must wear rubber gloves to prevent contact of the wastewater with the skin. Gloves should be thoroughly washed before removal and then hands should be washed using a disinfectant soap.

In many treatment plants, raw and treated water is piped directly into the laboratory, facilitating sampling. However, in other plants, the operator must take samples from open bodies of water which can be dangerous. Do not climb over or go beyond guardrails or chains when collecting samples. Instead, use sample poles, ropes, or other equipment to safely collect samples.

Sampling Locations

Operators take samples of raw water to determine water characteristics which will influence the treatment procedure, and then they take samples of finished water to determine how well the treatment worked.

Since you want the sample water to be representative of the water at a given location, you should always take samples from well-mixed areas of tanks or pipes. If you wanted to determine the quality of drinking water reaching the customer, it would be a bad idea to take a water sample from a dead end in the distribution system since water in such a location is unlikely to be representative of water in the distribution system as a whole. Similarly, if you are sampling water in the flocculation basin, you should not take your sample from the stagnant water at the very bottom of the basin which is not well-mixed.

Sampling Equipment

In mo the equipment used to collect samples is relatively simple and consists of sampling containers such as jars and beakers. Only a few tests, such as dissolved oxygen testing, require that specialized devices be used to collect samples. If the water to be sampled is out of reach, a long-handled aluminum dipper attached to a wooden pole can be used to collect a water sample.

Sampling containers must be made from a material which is corrosion-resistant and leak-proof, will withstand repeated refrigeration and cleaning, and will not interact with the sample water.

Sample bottles should be made of plastic (above) or glass (right). Each bottle should have a leak-proof lid.

In general, plastic or glass should be used for sample containers. Contamination of sample containers can be prevented by having a different collection container for each sampling location and by cleaning sample containers thoroughly after each use. Between each collection, the sample container should be cleaned with hot water and a good quality detergent, rinsed thoroughly with tap water followed by distilled water, and allowed to dry completely. At intervals, the sample containers should also be acid cleaned to remove all residues. Except in the case of samples for bacteriological or oil and grease tests, sample containers should be rinsed at the time of sample collection with a portion of the water to be sampled.

Sample Quality

Samples should be taken from locations in which the water is well-mixed and is representative of the body of water being sampled. In addition, the operator should avoid collecting large or unusual particles in routine samples. Samples which contain deposits, growths, or floating material which have accumulated at the sample point or on the side walls will tend to result in skewed test results, so these materials should be avoided.

Types of Samples:

There are two main types of samples which are used in water and wastewater treatment - grab samples and composite samples. The type of sample taken in a given instance will depend on the type of test to be performed, the frequency of testing, and on permit requirements. We will explain each test procedure below.

Grab Samples

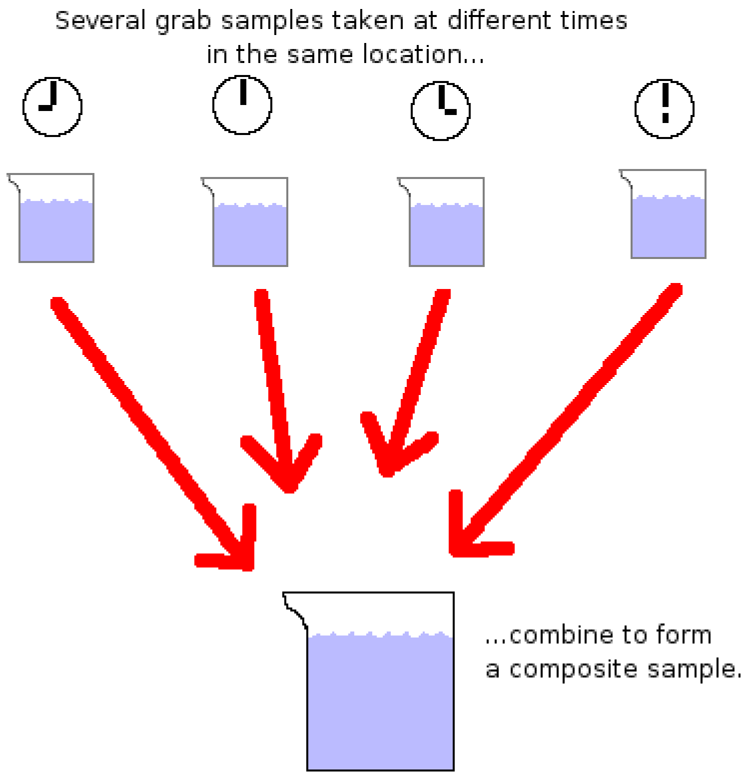
A grab sample, also known as a catch sample, consists of a single sample taken at a specific time. This is the most common type of sample and is the sampling technique you will use for most of your labs. For example, you took a grab sample when you collected a beaker of raw water and tested it for pH.

A grab sample has certain limitations. In essence, a grab sample takes a snapshot of the characteristics of the water at a specific point and time, so it may not be completely representative of the entire flow. Grab samples are most appropriate to small plants with low flows and limited staffs that cannot perform continual sampling. On the other hand, grab samples do provide an immediate sample, and are thus to be preferred for some tests.

Specifically, pH, dissolved oxygen, and total residual chlorine can change very rapidly in water once the sample is removed from the flow, so grab samples are preferred.

Grab samples must be collected carefully to make them as representative as possible of the water as a whole. They should be taken at a time of day when the plant is operating near its average daily flow rate. If grab samples are used to determine plant efficiency by collecting a raw water sample and a treated water sample, then the collection of the effluent should be delayed long enough after collection of the influent sample to allow for the raw water to pass completely through the treatment process. Finally, be aware that mixing two or more grab samples may not result in a result which averages the characteristics of the samples.

Composite Samples



A composite sample, also known as an integrated sample, is a sample which consists of a mixture of several individual grab samples collected at regular and specified time periods, each sample taken in proportion to the amount of flow at that time. Composite samples give a more representative sample of the characteristics of water at the plant over a longer period of time.

The greatest strength of composite samples is their ability to take into account changes in flow and other characteristics of the water over time. This helps the operator gain an overall picture of the total effects that the influent will have on the treatment process and that the effluent will have on the receiving water. However, composite samples cannot be used for tests of water characteristics which change during storage (such as dissolved gases) or of water characteristics which change when samples are mixed together (such as pH.)

Composite samples are often taken using automatic sampling devices. These may be set to take a sample every 8, 12, or 24 hours, with the frequency depending on test requirements, on the size of the treatment plant, and on permit requirements.

Sample Volumes for Composite Samples

One most important aspect of a composite sample is that each individual grab sample must be proportional to the amount of flow at the time the sample was collected. Most automatic equipment used to take composite samples will make these calculations for you and will collect a correctly sized grab sample during each time period.

The volume of sample collected at any given time depends on the volume of flow at that time, the total flow for the day, the total composite sample volume, and the number of individual grab samples to be taken. The following equation can be used to calculate a grab sample's volume:

$$\text{Grab sample volume, mL} = \frac{\text{Flow rate at sample time} \times \text{Composite sample volume, mL.}}{\text{Number of grab samples} \times \text{Average daily flow}}$$

Storage

The best test results will be achieved if you analyze all samples as soon after collection as possible. However, this is not always possible, so samples should be preserved.

Preservation

Except when collecting a grab sample and immediately testing it, storage is a nearly ubiquitous part of the sampling process. Composite samples must be stored carefully.

There are a variety of techniques which can be used to preserve samples during storage to Prevent their deterioration, including chemical addition and refrigeration. The type of chemical added to a sample will depend on the nature of the test to be performed, but most samples can benefit from refrigeration during storage. When refrigerating samples, keep the temperature at or below 4°C but do *not* allow the samples to freeze. At this temperature, biological activity is significantly reduced. However, you should be aware that no preservation technique completely stops biological and chemical action, so samples cannot be preserved for great lengths of time.

Pre-treatment is most necessary when dealing with wastewater samples with heavy solids concentrations or with different sized solids, such as that found in the wastewater influent or in mixed liquor. These samples should be homogenized in a blender before testing.

Answer the following questions:

Q-1	State the broad objectives of sampling.
Q-2	Differentiate grab and composite sampling.
Q-3	The average daily flow at a plant is 11.3 MLD and the total volume of composite sample is to be 4,000 mL made up of 24 grab samples. At the time of first sample, the plant's flow is 5.2 MLD, what will be the volume of grab sample?
Q-4	Enlist points to be considered in sample storage and preservation.

Marks Obtained:	**Signature of Faculty:**	**Date:**

Date:

2.

SAMPLING OF AMBIENT AIR FOR TOTAL SUSPENDED PARTICULATE MATTER WITH HIGH VOLUME SAMPLER

Objective: Measurement of total suspended particulate matter with high volume sampler.

High Volume Sampler:

High volume sampler is a basic instrument used primarily for measuring concentration of suspended particulate matter in atmospheric air. By definition, suspended particulates are too small in size to have an appreciable falling velocity and are likely to remain in the atmosphere for significant periods of time. These particulates usually range from 1 micron to approximately 100 microns in size and may be caused by a variety of processes such as incomplete combustion of solids, liquid, or gaseous fuels, wastes from metallurgical, chemical and refining operation, incineration, etc. Moreover, natural sources also contribute suspended materials like spores, salt water spray and pollens.

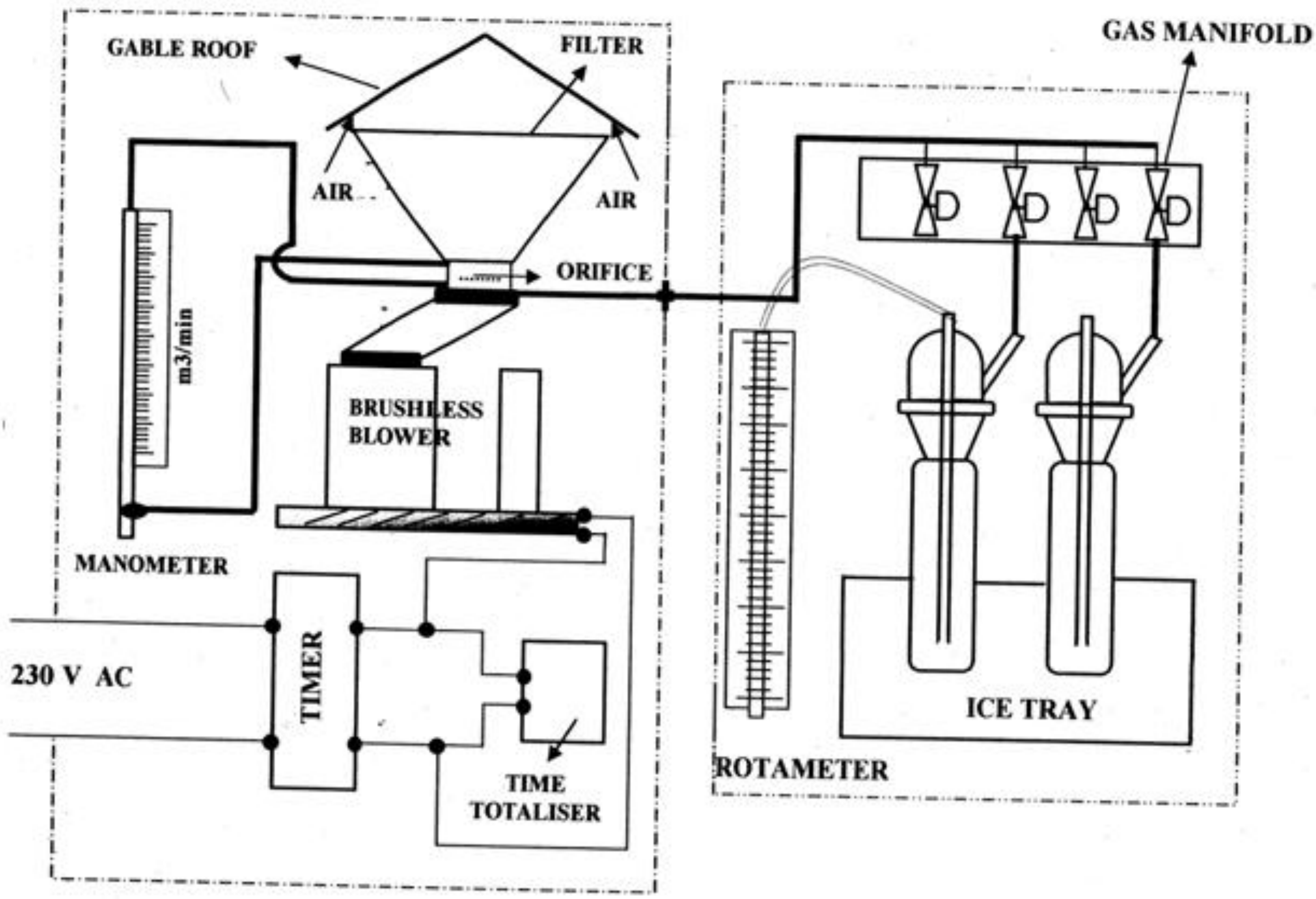
High volume sampling is an internationally accepted standard technique for monitoring the concentration of suspended particulates. In these systems a large volume (1500 cubic meters) of atmospheric air is passed through a suitable filter medium over a period of up to 24 hours. They will thus yield measuring dust samples in areas with dust levels as low as one microgram per cubic meter of air. However, where dust concentrations are high, shorter sampling times may suffice.

Apparatus:

1. Sampler: A sampler consisting of protective housing, blower, voltage stabilizer, time totalizer, rotameter/orifice meter, and filter holder capable of supporting a 20.3 cm x 25.4 cm glass fiber filter.
2. Volume flow controllers: Sampler flow rate should be maintained within 15% of the designed flow rate (1000 L/min.)
3. Analytical Balance: having a sensitivity of 0.01 mg.
4. Flow metering device: accurate to ±5%.
5. Equilibration rack: the rack to separate filters from one another so that the equilibration air can reach all parts of the filter surface.

6. Filter media: A 20.3 cm x 25.4 cm glass fiber filter.

7. Filter jacket: a smooth, heavy paper folder or envelope is used to protect the filter between the lab and field during storage. Filter and sampling data are often recorded on the outside of the jacket, but this should not be done while the filter is in the jacket to prevent damage.



SCHEMATIC DIAGRAM OF APM 430 AND APM 411

Procedure:

1. Calibration of sampler:

The sampler shall be periodically calibrated at least once in six months or whenever a major repair/replacement of blower takes place, by top loading calibrator of national standard.

2. Inspection of filter:

Prior to use, expose each filter to a light source and inspect for pinholes, particles, tears, creases, limps and other imperfections. Filter with visible defects should not be used. A small brush is often used to remove stray particles adhering to the surface of new filters. Always handle filter papers from their edges and do not crease or fold the filter medium prior to use.

3. Filter equilibration:

Place blank or exposed filters in air tight desiccators having active desiccant in the control temperature 15 to 27°C and 0-50% relative humidity environment prior to weighing. It is usual to put an identification number and date of sampling on the filters. Weigh the filters to the nearest milligram and record the weight and filter identification number.

4. Field sampling:

Selection of sampling site: The sampler is usually operated at ground level. In normal usage it is never operated more than 15 meters above ground level. In order to obtain a representative sample, the sampler should not be positioned near a wall or other obstructions that would prevent free air flow. In excessively turbulent condition or in the presence of strong surface winds or otherwise inclement weather, the sampling rate is likely to decrease rapidly and perhaps in a non-linear fashion due to filter choking.

Always install or remove the filter only when the sampler is OFF. Open the gable roof of the shelter, loosen the wing nuts and remove the face plate from the filter holder. Remove the filter from its jacket and centre it on the support screen with rough side of the filter facing upwards. Replace the face plate without disturbing the filter and fasten securely. Under tightening will allow air leakage, over tightening will damage the rubber face plate gasket. A vary light application of talcum powder may be used on the rubber gasket to prevent the filter from sticking. Close the roof of the shelter.

Connect the mains chord of the sampler to a live 220 V.A.C. outlet. Switch on the machine. Allow the Blower to run for a minute so that it attains full speed and then record the sampling rate indicated by the orifice meter. In case gases are also to be sampled, set the desired flow rates (for gaseous sampling) using the needle valves of the gas manifold. Record the sampling rates for gases.

After the sampling is over, record the final flow-rate indicated by the orifice meter. Record the flow-rate indicated by the rotameter (for gaseous sampling). In case the systems was operating under the control of the Timer and the Blower has already been shut off, restart the blower using the ON-OFF switch and allow the flow rate to stabilize for a minute before recording the flow rates mentioned above. Record the final sampling time indicating by the Time totalizer. Open the filter-holder and carefully remove the Filter Paper. Fold the paper along its length so that the soiled sides are in contact and are facing inwards. Store the filter paper in the jacket.

Calculation (For Suspended Particulates Sampling):

Weight of suspended particulates (W)

$W = W_2 - W_1$ (grams)

W_2 = Weight of the filter paper after sampling (grams)

W_1 = Weight of fresh filter paper (grams)

Volume of Air Sampled (V)

$V = Q \times T$ (Cubic Meters)

Q = Average sampling rate (Cubic meters per minute)

T = Sampling Time (Minutes)

$Q = (Q_1 + Q_2)/2$

Q_1 = Initial sampling rate indicated by the Orifice Meter at the start of sampling

Q_2 = Final Sampling rate indicated by the Orifice Meter just before the end of sampling.

Concentration of Suspended Particulate Matter = $W \times 10^6 / V$ (µg/cubic meter)

Environmental significance of suspended particulate matter measurement:

Conclusion:

Marks Obtained:	Signature of Faculty:	Date:

Date:

3.
SOLID WASTE SAMPLING AND ANALYSIS

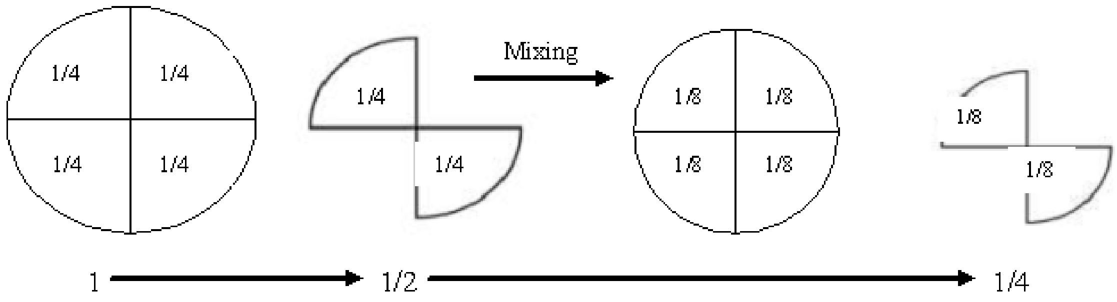
Objective: Sampling and characterization of solid waste from the campus.

Solid Waste:

Municipal Solid Waste (MSW) is the trash or garbage that is discarded day to day in a human settlement. According to MSW Rules 2000 MSW includes commercial and residential wastes generated in a municipal or notified areas in either solid or semisolid form excluding industrial hazardous wastes but including treated bio-medical wastes. Waste generation encompasses activities in which materials are identified as no longer being of value (being in the present form) and are either thrown away or gathered together for disposal. Municipal Solid Waste consists of the following kinds of waste.

Methods of Waste Characterization

Physical Characteristics of waste must be ascertained through '*quarter coning method*', as the data of waste composition is essential for evaluating feasible techniques for treatment.



Quarter coning method

Requirements of sampling:

It is of utmost importance for all analyses that the samples be representative.

Refuse

A mobile hammer mill shall be used to take a crude sample of refuse so that the refuse to be examined may be ground and homogenized on the spot.

Crude refuse 100 to 200 kg may be taken as the basic unit for a sample which may be taken at variable intervals depending on the amount of waste In accordance with the purpose of the analysis, 2 to 4 individual samples may be ground together in order to reduce the number of samples during more lengthy tests.

The ground refuse has to be mixed intensively. During mixing, about 10 to 20 small individual portions are taken and filled into airtight plastic bags in quantities of 1 to 2 kg, to be stored as samples. This heterogeneous mixture is the *crude sample.*

Compost

Grab samples (1 to 2 kg) shall be taken from as many parts of the windrow as possible and well mixed It shall be ensured that equal amounts are taken from all parts. The actual sample of 1 to 2 kg shall be taken from this mixture. In the same way compost samples shall be taken from the test material in plastic baskets and small windrows. The mixture obtained in this way is the *crude sample.*

Sewage Sludge

Before a sample is taken, the sewage sludge shall be well homogenized However, too intensive mixing with a high speed agitator should be avoided if the sample is to be used in dehydration tests (flocculation and filtration tests) The sewage sludge obtained In this way is the *crude sample.*

Residue After Incineration

The crude sample shall be taken directly from the conveying system under examination if possible. Fly ash and riddling shall be taken from the corresponding areas.

The size of the sample shall depend on the kind of information sought and the structure of the material. In the case of single samplings (which may have a random composition), a large quantity should be taken and prepared For longer examination periods, in which several Single Samples are taken at smaller intervals, the quantity may be fixed at 100 kg per sampling. Considerable deviations may occur with smaller quantities.

Procedure:

Take grab samples from different institute buildings, workshop, garden, laboratories, and canteen in dustbins given.

Mix all grab samples collected and take a representative sample by quartering it.

Segregate the waste based on the type of constituent present in it.

Observations:

Weight of sample = kg

Sr. No.	Material	Weight in kg	Weight in %

Environmental significance:

Conclusion:

Marks Obtained:	Signature of Faculty:	Date:

Date:

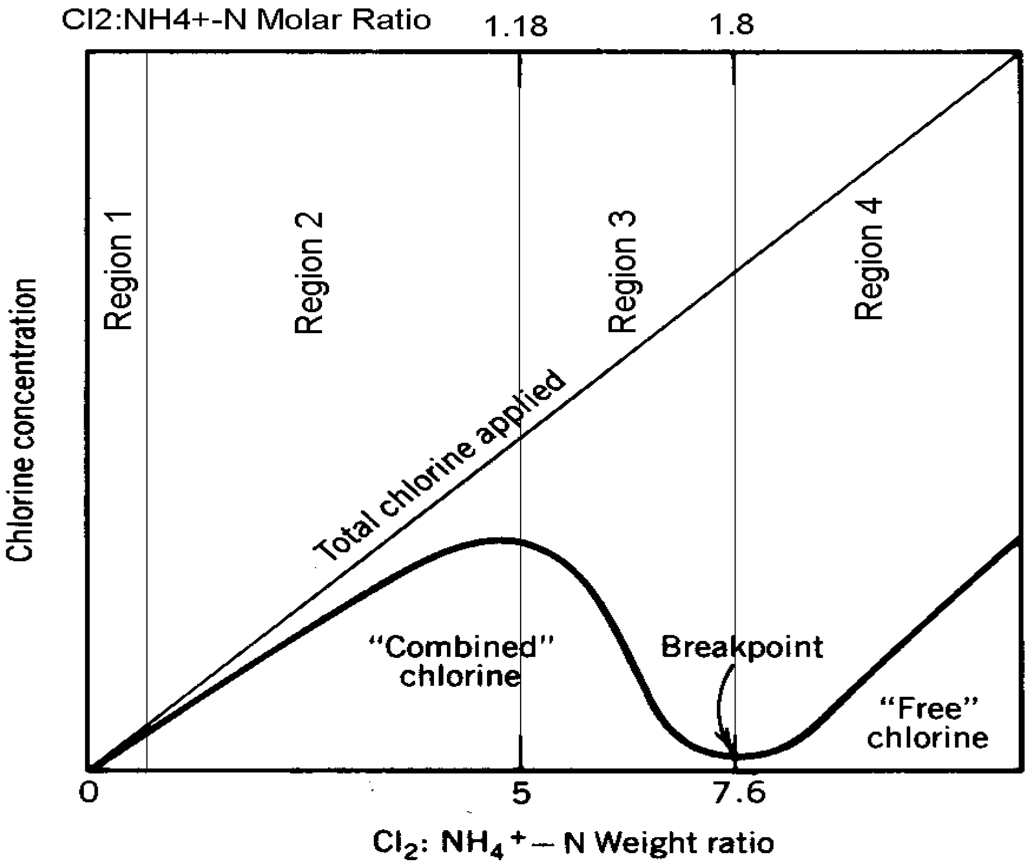
4.

BREAK POINT CHLORINATION

Objective: To determine the chlorine demand for given sample and break point chlorination

Chlorination: Water used for drinking and cooking should be free of pathogenic (disease causing) microorganisms that cause such illnesses as typhoid fever, dysentery, cholera, and gastroenteritis. Purification of drinking water containing pathogenic microorganisms requires specific treatment called disinfection. Several disinfection methods eliminate disease-causing microorganisms in water, chlorination is the most commonly used.

Principle: Chlorine combines with water to form hypochlorous and hypochloric acid. Hypochlorous acid dissociates to give the OCl^- and HOCl depend upon pH of the solution. Hypochlorides also give the OCl^- ions, HOCl rupture the cell membrane of microbes (disease producing organisms). These also react with the impurities like ammonia, oxidizable organic matter like ferrous ions, nitrites, etc. to form chloramines and stable ions of the latter respectively.



The "Breakpoint Chlorination" Curve: Reaction of HOCl with Pure H_2O + NH_3

in fig,

Regions 1 and 2: Formation of combined Cl_2 dominates over oxidation of N; concentration of residual Cl_2 increases with increasing Cl_2:N ratio

Region 3: Oxidation of N dominates over formation of combined Cl_2; when Cl_2:N ratio increases, more combined Cl_2 is oxidized than is formed

Region 4: All N oxidized; increase in Cl_2:N ratio leads to increase in free Cl_2 residual

Apparatus: Beakers, conical flasks, burette

Reagents: Standard chlorine solution, acetic acid, potassium iodide, 0.01 N standard sodium thiosulphate solution, starch indicator.

Procedure: :

1. Take 1 liter of samples in six jars and add varying amount (1, 2, 3, 4, 5, 6, 8 mg/l) chlorine solution to each jar.

2. Stir the sample with a glass rod/mechanical mixer for complete mixing. Allow 20 minutes contact time for chemical reaction.

3. Take 25 ml of sample in a conical flask and add 1ml of acetic acid and 1 gram of potassium iodide and allow the reaction to complete.

4. Titrate the sample with 0.01 N standard sodium thio-sulphate solution until the yellow color of the librated iodine is almost faded out.

5. Add 1 ml of starch indicator, the solution turns blue. Continue the titration till the blue color disappears. Note down the volume of sodium thio-sulphate consumed (A).

6. Repeat the procedure for blank sample (B).

7. Determine the available residual chlorine in the samples.

8. Plot graph between concentration of chlorine added and residual chlorine to find out the break point chlorination.

Observations and Calculations:

Chlorine dosage (mg/l)	**Residual chlorine(mg/l)**

Result
Break point chlorination: …………………..

Conclusion:

Environmental Significance:

Marks Obtained:	Signature of Faculty:	Date:

Date:

5.

DETERMINATION OF CO-EFFICIENT OF AERATION

Objective: To determine the coefficient of aeration (K_{La}) by dynamic method

Introduction:

Oxygen transfer, the process by which oxygen is transferred from the gaseous state to the liquid phase is a vital part for number of wastewater treatment processes. Oxygen acts a limiting nutrient for an aerobically growing culture in the bioreactor. The functioning of aerobic processes depends on the availability of sufficient quantities of oxygen. Therefore oxygen has to be adequately supplied to enhance the dissolved oxygen so that microorganisms will be able to survive. It is important to note that the capacity of water to retain oxygen is rather limited. Under the normal condition of 25º C and 1 Atmosphere pressure, the dissolved oxygen will be 8 mg/L. Air has to be continuously purged and agitated so as to enhance the depleting oxygen in the reactor. For adequately aerated cultivation, the supply of oxygen has to be greater than the demand of the culture to maintain the healthy growth & metabolite production by the microorganisms at any point of time. Assuming that the reactor is in well mixed condition and dissolved oxygen concentration is constant through-out the reactor, the following equation will hold true:

$$dC/dt = K_{La} (C_s - C_t)$$

dC/dt – Rate of Oxygen transfer, mg/L-h

K_{La} – Mass transfer coefficient, time^{-1}

C_s – Saturation dissolved oxygen concentration, mg/L

C_t – Dissolved oxygen concentration in liquid, mg/L

Apparatus:

1. Diffused Aerator
2. DO Meter

Reagents:

1. Sodium Sulphite

Procedure:

1. Take 1L of water sample and deoxygenate the sample using sodium sulphite.
2. Calculate the amount of sodium sulphite required to bring dissolved oxygen content to zero.

3. After the DO content is brought to zero, immerse diffused aerators in it.

4. Calculate the amount of DO at different time intervals as given in the table.

5. Obtain the value of C_s as per the temperature of water sample.

5. Plot the graph of ln (C_s- C_t) versus time to attain the value of K_{La}.

The oxygen saturation concentrations in the water as a function of water temperature:

t °C	mg O_2/l	t °C	mg O_2/l	t °C	mg O_2/l
0	14.65				
0.5	14.45	10.5	11.14	20.5	8.93
1	14.25	11	11.00	21	8.84
1.5	14.05	11.5	10.87	21.5	8.76
2	13.86	12	10.75	22	8.67
2.5	13.68	12.5	10.62	22.5	8.58
3	13.49	13	10.60	23	8.50
3-5	13.31	13.5	10.38	23.5	8.42
4	13.13	14	10.26	24	8.33
4.5	12.96	14.5	10.15	24.5	8.25
5	12.70	15	10.03	25	8.18
5-5	12.62	15.5	9.92	25.5	8.10
6	12.46	16	9.82	26	8.02
6.5	12.30	16.5	9.71	26.5	7.95
7	12.14	17	9.61	27	7.87
7.5	11.99	17.5	9.50	27.6	7.80
8	11.84	18	9.40	28	7.72
8.5	11.70	18.5	9.30	28.5	7.05
9	11.55	18	9.21	29	7.58
9.5	11.41	19.5	9.12	29.5	7.51
10	11.27	20	9.02	30	7.44

Observation Table:

Time (min)	DO, C_t (mg/L)	ln (C_s- C_t)
0		
1		
2		
3		
4		
5		
6		
7		
8		

10		
12		
15		
20		
25		
30		
35		
40		
45		
50		
55		
60		

Conclusion:

Environmental Significance:

Marks Obtained:	**Signature of Faculty:**	**Date:**

6.
DETERMINATION OF COLOR REMOVAL BY ADSORPTION

Objective : To determine the efficiency of color removal by activated carbon as adsorbent

Introduction

Adsorption involves the inter-phase accumulation of concentration of substances at a surface or interface. Adsorption, the binding of molecules or particles to a surface, must be distinguished from *absorption*, the filling of pores in a solid. The binding to the surface is usually weak and reversible. Just about anything including the fluid that dissolves or suspends the material of interest is bound, but compounds with color and those that have taste or odor tend to bind strongly. Compounds that contain chromogenic groups (atomic arrangements that vibrate at frequencies in the visible spectrum) very often are strongly adsorbed on activated carbon. Decolorization can be wonderfully efficient by adsorption and with negligible loss of other materials.

The most common industrial adsorbents are activated carbon, silica gel, and alumina, because they present enormous surface areas per unit weight. Activated carbon is produced by roasting organic material to decompose it to granules of carbon - coconut shell, wood, and bone are common sources. Silica gel is a matrix of hydrated silicon dioxide. Alumina is mined or precipitated aluminum oxide and hydroxide. Although activated carbon is a magnificent material for adsorption, its black color persists and adds a grey tinge if even trace amounts are left after treatment; however filter materials with fine pores remove carbon quite well.

A surface already heavily contaminated by adsorbates is not likely to have much capacity for additional binding. Freshly prepared activated carbon has a clean surface. Charcoal made from roasting wood differs from activated carbon in that its surface is contaminated by other products, but further heating will drive off these compounds to produce a surface with high adsorptive capacity. Although the carbon atoms and linked carbons are most important for adsorption, the mineral structure contributes to shape and to mechanical strength. Spent activated carbon is regenerated by roasting, but the thermal expansion and contraction eventually disintegrate the structure so some carbon is lost or oxidized.

Apparatus

1. Conical flasks
2. Magnetic Stirrer
3. UV-VIS Spectrophotometer

Reagents

1. Granular Activated Carbon
2. Methylene blue dye

Procedure

A) Adsorption Kinetics

1. Make standards from 10^{-5} M solution of methylene blue dye at concentrations of 5.0 x 10^{-8}, 1.0x 10^{-7}, 1.5 x 10^{-7}, 2 x 10^{-7}, 2.5 x 10^{-7}, 3 x 10^{-7}, 4 x 10^{-7} and 5 x 10^{-7} moles/L and prepare the calibration graph of concentration of Methylene blue (MB) versus absorbance taken at wavelength of 661nm.
2. Prepare conical flasks with 0.2 g of activated carbon at initial dye concentration of 5 x 10^{-7} moles/L and keep it on magnetic stirrer at 150 rpm.
3. Take the supernatant from sample at 5,10,15,20,25,30,40,60,75,90 and 120 minutes.
4. Analyze the samples for residual color concentration with the help of calibration curve.

B) Adsorption Equilibrium

1. Prepare conical flasks with different amounts of activated carbon at initial dye concentration of 5 x 10^{-7} moles/L and keep it on magnetic stirrer at 150 rpm.
2. Calculate the amount of MB dye adsorbed per unit weight of Activated Carbon (mg/mg).

C) Isotherm models

1. Prepare Langmuir and Freundlich Isotherms respectively and show which isotherm suits the best.

Observation Table

1. Preparation of Calibration curve

Sr.No	Concentration of MB (moles/L)	Concentration of MB (mg/L)	Absorbance
1	5.0×10^{-8}		
2	1.0×10^{-7}		
3	1.5×10^{-7}		
4	2.0×10^{-7}		
5	2.5×10^{-7}		
6	3.0×10^{-7}		
7	4.0×10^{-7}		
8	5.0×10^{-7}		

2. Adsorption Kinetics

Time (min)	Absorbance	Concentration of MB (mg/L)	C/C_o	Moles of MB/gm of AC
5				
10				
15				
20				
25				
30				
40				
60				
75				
90				
120				

3. Adsorption Equilibrium

Wt. of AC (mg)	Absorbance	Concentration of MB (mg/L)	MB adsorbed per unit wt. of AC (mg/mg)
0			
50			
75			
150			
200			
250			
300			
350			
400			
450			
500			

Conclusion:

Environmental Significance:

Marks Obtained:	Signature of Faculty:	Date:

Date:

7.

DETERMINATION OF BOD RATE CONSTANT

Objective: To determine BOD reaction rate constant

Introduction

The determination of the BOD rate constant is important for understanding the nature of the wastewater. In case of discharging this wastewater into streams, the rate constant would help in predicting the impacts of discharging the wastewater: on the life in the stream, on the dissolved oxygen values in the stream, and on the BOD values in the stream. The BOD reaction rate is dependent on 3 factors such as nature of waste, the ability of the organisms in the system to utilize the waste and temperature. For complex waste, like sewage, the BOD rate constant depends upon the relative proportions of the various components. The BOD rate constant is high for the raw sewage (K (base e) = 0.35 - 0.7 per day) and low for the treated sewage (K (base e) = 0.12 - 0.23 per day), owing to the fact that, during wastewater treatment the easily degradable organic matter will get more completely removed than the less degradable organics. Hence, in the treated wastewater, relative proportion of the less biodegradable organic matter will be higher, giving lower BOD rate constant. Many organic matters can only be utilized by particular group of microorganisms. In natural environment, where the water course is receiving particular organic compound, the microorganisms which have capability to degrade that organic matter will grow in predominant. The biochemical reactions are temperature dependent and the activity of the microorganism increases with the increase in temperature up to certain value, and drop with decrease in temperature. Since, the oxygen utilization in BOD test is caused by microbial metabolism, the rate of utilization is similarly affected by the temperature. To know the impact of wastewater on receiving streams determination of 5 day BOD is not enough. The required information can be obtained from BOD_u and reaction rate constant k. The rate constant depends on the type of wastewater and the presence of inhibitory compounds in it. Many methods are used for the calculation of BOD_u and k out of which Fujimoto method and least squares method are of prime importance. The Fujimoto's method is an arithmetic plot of BODn+1 against BOD . The value at the intersection of the plot n with a line of slope 1 corresponds to the ultimate BOD and the rate of BOD removal is determined. In the linear regression (least squares) method fitted a curve through a set of data points, so that the sum of the squares of the residuals must be a minimum.

Apparatus

1. BOD bottles
2. BOD incubator
3. Conical flasks
4. Burette
5. Pipette

Reagents

1. $MnSO_4$ solution
2. Alkali Iodide (KI) solution
3. Sodium thiosulphate (0.025N)
4. Starch Indicator

Procedure

1. Prepare respective reagents for DO determination.
2. Prepare dilution water using tap water by aerating water samples up to 2 hours. Use the same dilution water in BOD experiment.
3. Determine DO content for 1, 2, 3, 5, 6, 8 and 10 days with dilution factor of 50.
4. After obtaining DO values, obtain BOD_u and k using Fujimoto and Least squares method and plot graphs of BOD versus time.

Observation Table

Day	**Blank**		**Sample**		**BOD (mg/L)**
	DO_i	**DO_f**	**DO_i**	**DO_f**	
0					
1					
2					
3					
5					
6					
8					
10					

Conclusion:

Environmental Significance:

Marks Obtained:	Signature of Faculty:	Date:

Date:

8.

HEAVY METAL REMOVAL BY ADSORPTION

Objective: To remove metal like copper by optimizing pH, contact time and adsorbent dose using Activated Carbon.

Introduction

Pollution from heavy metals is a major concern in developing countries. The discharge of heavy metals into water-courses is a serious pollution problem which may affect the quality of water supply. Effluents from various processing industries such as electroplating industries is reported to contain high amounts of heavy metal ions, such as nickel, iron, lead, zinc, chromium, cadmium and copper. The presence of these heavy metals in industrial wastewaters is of serious concern because they are highly toxic, non-biodegradable, carcinogen, and continuous deposition into receiving lakes, streams and other water sources within the vicinity causes bioaccumulation in the living organisms. These perhaps, could lead to several health problems like cancer, kidney failure, metabolic acidosis, oral ulcer, renal failure and many more.

Iron (Fe^{2+}), lead (Pb^{2+}), and copper (Cu^{2+}) have been introduced into natural waters from various sources, such as wastewater discharged from hospitals and different industries, including smelting, metal plating, Cd–Ni battery, alloy manufacturing, phosphate fertilizer, mining pigment, and stabilizer production.

Physical and chemical treatment processes, such as coagulation, flotation, chemical precipitation, membrane filtration, and electrochemical methods, are used to remove heavy metals from wastewater. However, these methods have lack of significance mainly due to their high capital and operational cost, disposal of residual metal sludge, continuous need of chemicals, and, sometimes, failure to meet acceptable limits of environmental protection agencies. Therefore, there is a pressing need for more cost-effective and environmentally friendly methods. One of the most commonly used techniques involves the process of adsorption, which is the adhesion of chemicals onto a solid surface. Removal of heavy metal pollutants at high concentrations from water can be readily accomplished by chemical precipitation or electrochemical methods. At low concentrations, removal of such pollutants is more effective by ion-exchange or adsorption on solid sorbents such as activated carbon. At present, adsorption is widely accepted in environmental treatment applications throughout the world.

It is recommended that the absorbent is available in large quantities, of free or very low cost and easily re-generable. In the midst of a large variety of adsorbent available, activated carbon is the most important and cheapest adsorbent used in the current method of pollution control. Liquid–solid adsorption systems are based on the ability of certain solids to preferentially concentrate specific substances from solutions onto their surfaces. This principle can be used for the removal of pollutants, such as metal ions and organic compounds from wastewater. The presence of heavy metals in the environment is a major concern due to their toxicity for many life forms. Unlike organic pollutants, the majority of which are susceptible to biological degradation, heavy metals will not degrade into harmless end products. Therefore, the elimination of heavy metals from water and waste water is important to protect public health. Thus, treatment of aqueous wastes containing soluble heavy metals requires concentration of the metals into a smaller volume followed by recovery or secure disposal.

Apparatus

1. Conical flasks
2. Magnetic Stirrer
3. UV-VIS Spectrophotometer

Reagents

1. Granular Activated Carbon
2. Cuprethol Reagent

Prepare solution 1 by dissolving 2 g of diethanolamine in 100 ml of Methy alcohol and solution 2 by dissolving 1.5 ml of Carbon disulphide in 100 ml of Methy alcohol. Prepare the cuprethol reagent by mixing equal volumes of two solutions immediately after preparation.

3. Stock Copper solution
4. Sulphuric acid and NaOH

Procedure

1) Preparation of Calibration Curve

1. Make standards from copper stock solution (1g/L) of 0, 2, 4, 6, 8 and 10 mg/L copper concentration and make the volume to 10 ml using distilled water.
2. Add 0.1 ml of Cuprethol reagent, mix well and allow to stand for 10 minutes. Using distilled water as blank, read the concentration of copper at 440 nm.
3. Prepare a calibration curve.

4. Determine copper concentration of experimental samples using the calibration curve and the steps outlined in 1 and 2.

2) Optimization of pH

1. Take 10 flasks with adsorbent dose of 0.5g with initial concentration of 10 mg/L and vary pH values from 1-10.
2. Keep the flask for stirring at 150 rpm for 1 hour.
3. Take the supernatant and check the copper removal using calibration curve.
4. Draw the graph of pH versus Copper concentration and find the value of optimum pH.

3) Optimization of Adsorbent dose

1. Take 6 flasks and add different dosages of activated carbon with initial concentration of 10 mg/L.
2. Keep the flask for stirring at 150 rpm for 1 hour.
3. Take the supernatant and check the copper removal using calibration curve.
4. Draw the graph of adsorbent dose versus Copper concentration and find the value of optimum adsorbent dosage.
5. Cross check the copper removal at optimum pH and optimum adsorbent dosage.
6. Prepare adsorption isotherms and see which isotherm model suits the best.

Observation Table

1. Preparation of Calibration Curve

Sr.No	Copper concentration (mg/L)	Absorbance
1	0	
2	1	
3	2	
4	3	
5	4	
6	5	
7	6	
8	7	
9	8	
10	9	
11	10	

2. Optimization of pH

pH	Absorbance	Copper concentration (mg/L)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

3) Optimization of Adsorbent Dosage

Sr.No	Adsorbent Dosage (mg/L)	Absorbance	Copper concentration (mg/L)
1	0		
2	25		
3	50		
4	100		
5	200		
6	300		
7	400		
8	500		
9	800		
10	1000		

Conclusion:

Environmental Significance:

Marks Obtained:	Signature of Faculty:	Date:

Date:

8.
SLUDGE VOLUME INDEX

Objective: Determination of mixed liquor suspended solids, mixed liquor volatile solids & sludge volume index.

MLSS, MLVSS & SVI are important parameters measured regularly for operation of an activated sludge plant. While MLSS gives indication of total suspended solids resent in aeration tank, the MLVSS gives indication of dead & live biomass, SVI indicates vol. of settled sludge per gram of suspended solids. In an activated swage plan SVI may be co-related with the performance of the plant.

Principle:

- A well mixed sample of mixed liquor is filtered through a pre-weighted glass fiber or ash less fiber & residue retained on the filtered is dried to a const. weight at 103-105°c the increase in weight of the filter represents the MLSS.
- The residue along with the filter is ignited to const. weight at 550°c the remaining solids represents the fixed or inorganic solids present in MLSS. The difference bet. MLSS & inorganic / fixed solids give value of MLVSS.
- A well mixed sample of mixed liquor is poured quickly in 1 liter graduated cylinder vol. of sludge settled at the and of 30 minutes is recorded, SVI is calculated as,

$$SVI = \frac{\text{vol. of sludge settled in 30 min. (ml)}}{\text{MLSS in (gm/liter)}}$$

Apparatus:

Drying oven
Analytical Balance
One liter graduated cylinder
Filtering flash
Buckner funnel
Vacuum pump

Procedure:

Preparation of filter

Dry the filter in an oven at 103˙ – 105°C for one hour. Remove the filter from oven & cool it is desiccator, Repeat the cycle of drying, cooing & weighing until its const. weight is obtained or until weight change is less than of previous weight or 0.5 mg whichever is less.

Selection of filter & sample size

Choose sample Volume to yield between 2.5-5 mg dried residues. If the volume filters fails to meet the minimum yield increase sample volume as per requirement.

Sample Analysis

As sample filtering apparatus & filters 7 begin suction. Weigh the filters with a small volume of distill water to seat it properly in a funnel. Stir the sample of mixed liquor with a magnetic stirrer to obtain uniform suspension of solids or particles while string prepare a measured volume on to the filter. When required sample volume its is prepared wash the filter with 3-succesive 10 ml volume distill water allowing complete drainage between washing & continue suction for sample for about 3 minutes after filtration is complete samples with high dissolved solids may require additional washes carefully remove filter from filtration apparatus & transfer to a glass dish as a supports dry for at least one hour at 103^0-104^0 C in an oven, cool in a desiccators to ambient temp. & weight.

Calculation of MLSS:

$$\text{MLSS in ma/lit} = \frac{(a\text{-}b)\ \text{X1000}}{\text{Sample volume}}$$

Where, a= Weight of filter +dried residue in mg

b=initial weight of filter in mg.

Take filter & dry residue in a pre weigh porcelain, quartz crucible & ignite it at 550^0c in muffle furnace. Let the crucible cool partially in air until most of the heat has been dissipated. Weigh the crucible as soon as it has cooled to ambient temp.

Calculation of MLVSS:

$$\text{MLVSS in (mg/lit)} = \frac{\text{(c-d) X1000}}{\text{Sample vol.}}$$

Where,

c=initial weight of crucible filter +dried residue.

d= weight of crucible initial weight of filter +inorganic residue.

Pour a well-mixed sample of liquor quickly in one liter graduated cylinder. Let the sample settle till 30 min. record the vol. of sludge settled.

Calculation OF SVI

$$\text{SVI=} \frac{\text{vol. of sludge settled in 30min. (mg/lit)}}{\text{MLSS (mg/lit)}}$$

Observations:

Parameter	Influent	Effluent after aeration		
		24 hrs.	48 hrs	72 hrs
SVI				
MLSS				
MLVSS				

Conclusion:

Environmental significance:

Marks Obtained:	Signature of Faculty:	Date: